Quantum Mechanics II — Exercise Sheet 8

Oscillator states and related models

Variant: Questions only

Exercise 1 (A coherent superposition with even number). For $\alpha \neq 0$ define $\Phi = C(\psi_\alpha + \psi_{-\alpha})$, where $C > 0$ normalizes the vector. Using the coherent-state expansion, compute C, all number probabilities and $\langle H \rangle$. Give the limit as $\alpha \to 0$.

Exercise 2 (A bounded perturbation of two levels). Let H be the oscillator Hamiltonian and let

$$B = \lambda(|h_0\rangle\langle h_2| + |h_2\rangle\langle h_0|), \qquad \lambda \in \mathbb{R}.$$

Diagonalize $H+B$ on $\mathrm{span}\{h_0, h_2\}$ and describe its action on the other number states. Explain why the domain is $D(H)$ and why this is an exact finite-dimensional reduction.

Exercise 3 (Landau levels from an oscillator). For a particle of nonzero charge e in a nonzero constant magnetic field B, take as given that a partial Fourier transform in y reduces the planar Hamiltonian to

$$H_k = \frac{P_x^2}{2m} + \frac{(\hbar k - eBx)^2}{2m}, \qquad k \in \mathbb{R}.$$

(i) Find its oscillator centre, frequency and levels. (ii) Explain why the levels do not depend on k and why a fixed-k plane wave is not a normalized state on the whole plane.

Exercise 4 (A coherent-state projection threshold). Recall the coherent-state overlap $|\langle \psi_\alpha, \psi_\beta \rangle|^2 = e^{-|\alpha-\beta|^2}$. For $\beta = \mathrm{i}\alpha$ find the probability of the rank-one projection onto ψ_β in state ψ_α. (i) Find the smallest $|\alpha|$ for which this probability is at most $1/10$. (ii) Compare with the projection onto $\psi_{-\alpha}$ at this same value of $|\alpha|$.